

Isolation, characterization and multiplex PCR development of Bean Goose (*Anser fabalis*) microsatellite loci

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Abstract DNA from non-invasively collected samples may be used to identify individuals for monitoring birds. Here we describe 36 novel microsatellite loci from Bean Goose (*Anser fabalis*) identified using a next-generation sequencing approach. From 34 variable loci, we selected 12 tetranucleotide loci with short amplicon sizes, combined these into two multiplex PCR sets, and included a sex-typing marker. In 31 Bean Geese from a population in central Norway, we found 4–10 alleles per locus and observed heterozygosity ranging from 0.58 to 0.87. The combined probability of identity for the 12 loci was 6.5×10^{-13} . These novel microsatellite loci and their multiplex PCR assays will be useful for genetic analyses of the Bean Goose, including typing of non-invasive samples such as molted feathers. Cross-species application of the two multiplex PCR assays revealed that all 12 loci amplified successfully in four other *Anser* species.

Keywords *Anser fabalis* · Microsatellite loci · Multiplex PCR · Non-invasive genetic monitoring

Zusammenfassung

Isolierung, Charakterisierung und Entwicklung einer Multiplex-PCR Methode für Mikrosatellitenloci von Saatgänsen (*Anser fabalis*)

Für die Identifizierung von Individuen kann DNA aus nicht-invasiv gesammelten Proben zum Monitoring von Vögeln genutzt werden. In dieser Studie beschreiben wir 36 neue Mikrosatelliten-Genorte von Saatgänsen (*Anser fabalis*), die im Rahmen einer Next-Generation Sequenzierungsmethode identifiziert wurden. Aus 34 variablen Stellen selektierten wir 12 Tetranucleotidloci mit kurzen Amplikonlängen, kombinierten diese in zwei Multiplex-PCR Sets und fügten einen Geschlechtsbestimmungsmarker hinzu. Bei 31 Saatgänsen aus einer Population in Zentralnorwegen fanden wir 4–10 Allele pro Ort und stellten Heterozygositäten von 0.58–0.87 fest. Die kombinierte Wahrscheinlichkeit der Identität für die 12 Orte war 6.5×10^{-13} . Diese neuen Mikrosatellitenorte und ihre Multiplex-PCR Methoden einschließlich der nicht-invasiven Probennahme wie beispielsweise Mauserfedern, können nützlich für genetische Analysen bei Saatgänsen sein. Die Anwendung der zwei Multiplex-PCR Methoden auf andere Arten zeigten, dass sich alle 12 Genorte erfolgreich bei vier anderen *Anser*-Arten amplifizierten.

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Feathers, fecal pellets, and other non-invasively collected materials may provide an easily available and reliable source of avian DNA (Segelbacher 2002; Horváth et al. 2005; Jacob et al. 2010), which can be used, for instance, to identify individuals and to estimate population size and adult turnover rate (Rudnick et al. 2005; Schwartz et al. 2007). Non-invasive genetic monitoring may thus represent

Table 1 Characterization of 36 novel Bean Goose (*Anser fabalis*) microsatellite loci

Locus	Primer sequences (5'–3')	Primer <i>T_m</i> (°C)	Repeat motif	<i>n</i>	<i>A</i>	Expected allele size (bp)	Observed allele size range (bp)	GenBank no.
Afa01	F: TAAAGAGCAGAACGCAACAG R: TAATGGCAACTGTCCAAGGC	57.1 59.2	(TATC) ₁₂	7	4	236	225–238	KT698169
Afa02	F: CGAATCACGAGGCACAAGAG R: TCCATTTGTAGGTGAGAAAATATGC	59.7 58.4	(TATC) ₁₁	5	6	209	196–216	KT698170
Afa03	F: AAGCCTTCCTTCATGTGAGC R: AGACAGTGAGATGCCCAGTC	58.9 59.9	(AAAC) ₁₀	7	3	199	184–195	KT698171
Afa04	F: AAAAGGCCCTGGGACCATAG R: AGGAAACCTGAAATGACACGG	60.2 59.3	(ATCT) ₁₂	7	4	244	239–255	KT698172
Afa05	F: GCATTTAGAGAACGGATACAGGG R: TGCCATGCAGAATAGAAAGGC	59.7 59.7	(TATC) ₁₁	7	4	192	177–198	KT698173
Afa06	F: CTGTCCATGAGTGGTAAGCC R: AGAACTGACTGTTGTGCATGTC	58.7 59.9	(AAAC) ₇	6	2	215	203–211	KT698174
Afa07	F: TCCTGTGCTGGATGGACAAC R: GGCATACAGGTCTGAGGAGG	60.7 60.0	(AAAC) ₈	7	2	225	218–222	KT698175
Afa08	F: TGTCTCTAAAGCAACACCTG R: ACTACCTGAGGCACTTGTTTTC	58.9 59.3	(AAT) ₇	7	2	199	188–194	KT698176
Afa09	F: TGGTAGATAAATACACTTGTGGC R: GCTCTGGGTTTAGAGTTTCGTC	57.1 59.8	(AAAC) ₁₀	5	3	220	207–215	KT698177
Afa10	F: ACTTTGTAGCAATAATGAAACTACGC R: GCTCCCCTTCCCATAGTCG	59.5 60.0	(AAAG) ₁₉	5	4	224	217–229	KT698178
Afa11	F: TTGAACGGTGATTGTGGCAG R: AGGCTGTTCGTACATTCAAGAC	59.8 59.4	(TTGT) ₇	7	2	215	207–215	KT698179
Afa12	F: TTCTGTTACAAACGGCTGGC R: TGTTTTATTGGACCTTCCCCC	59.8 58.6	(ACAA) ₇	7	2	249	245–249	KT698180
Afa13	F: TCTTATCTCCAAACAAGCTGGG R: GGTAGGAGGTGCTATAGGCG	58.8 59.9	(TATC) ₁₁	7	5	220	206–226	KT698181
Afa14	F: TTTCTGGGTTACCTGAAGCG R: AGAGTACACCTCATGCTCTG	58.6 57.1	(AAT) ₇	7	1	238	237	KT698182
Afa15	F: CATGTCAACAGATAGGCCTGC R: CCCAACGAAGTGCTGAATGG	59.8 60.2	(TCTT) ₇	7	3	128	225–242	KT698183
Afa16	F: TTCCCAGGTGTGCTTACTAC R: GCAGGACCCTGTTCTTTTAC	57.6 59.6	(AAAC) ₈	7	1	245	241	KT698184
Afa17	F: CAAATCAGGCCACACTGAC R: TGCCCTCTTCATCCTTAGCG	59.9 60.3	(ATAG) ₁₂	7	6	247	220–248	KT698185
Afa18	F: ACCCTCTATACCCTCTGGGC R: TCCTCAGTCATTTCTTTAACACAC	60.6 57.9	(TCTA) ₈	7	5	187	178–198	KT698186
Afa19	F: AGAAACCCAAATTTTATGTTGTTGTC R: GCTACTGCCCATTTCCACAG	58.9 59.7	(TTTG) ₇	7	2	175	173–177	KT698187
Afa20	F: AAGCGATCCTTCCTCTCCAC R: ACAGGGAGTCTGAGATAGTCAAG	59.9 59.5	(TCTA) ₁₆	7	5	188	187–203	KT698188
Afa21	F: AGTTGGAGGACTAAGGACACG R: GACCTCCAGCAAAATGGACAG	59.9 59.9	(AAAC) ₈	7	2	151	139–147	KT698189
Afa22	F: GGTCACAGGTTCTAGCTTTCC R: TTGGCTAAGGTAGGTGAGGG	59.1 59.2	(GATA) ₁₀	7	5	207	185–221	KT698190
Afa23	F: GTGTCTGATGCACAGTAGCC R: TGTCTTGGTCTCAGTAGCAC	59.4 60.2	(TATC) ₇	7	3	208	205–226	KT698191

Table 1 continued

Locus	Primer sequences (5′–3′)	Primer T_m (°C)	Repeat motif	n	A	Expected allele size (bp)	Observed allele size range (bp)	GenBank no.
Afa24	F: TGCCTCCTGCTGTAGATGAC R: TGAGCTTCACGTTGCTGAGG	59.9 61.4	(AAC) ₇	7	4	249	242–256	KT698192
Afa25	F: TGCATCAGTGATACCATTTGAG R: ATGCTGAGGCAACTCTACCC	57.0 60.2	(ATGT) ₁₁	7	5	246	228–244	KT698193
Afa26	F: ACTATGGACAGGCTGGGTTG R: CTGCCCCGTTAGAATACGTCC	60.2 58.9	(GATA) ₇	7	2	229	222–226	KT698194
Afa27	F: ATCTGGGTGCCTAGAGTTGC R: TCAGTGGAACGTGTTTGTGC	60.2 60.0	(AAAG) ₉	7	5	234	206–229	KT698195
Afa28	F: CGCCAAAACATCTCTGGAC R: AGGGAGGTCATACTGCAACC	59.9 59.9	(ATAG) ₁₁	7	3	246	239–247	KT698196
Afa29	F: ACGCATGATTCAATGGCTGG R: ACTGAAACCTGCACCTACCC	60.0 60.4	(ATAG) ₉	7	2	244	239–243	KT698197
Afa30	F: GTCTTGCACTTCACTCACCAG R: GAGACTGGCAGCCAGAAAAAG	59.8 59.9	(AGAT) ₁₀	7	6	129	125–150	KT698198
Afa31	F: TGAAGTAAACGGGCTTGTGATAG R: TCCTCCCCAAAATGCAAAGC	59.5 59.8	(AGAT) ₉	7	4	247	242–258	KT698199
Afa32	F: TGCCAGCAACAAAAATGTC R: CAGTGAACATGACTGATGGGG	58.5 59.4	(TATC) ₁₀	7	5	230	228–244	KT698200
Afa33	F: TGATTATGCAACGCCGTCC R: TACGTACAGACCATGGAGCC	60.0 59.7	(AAAC) ₇	7	3	103	89–101	KT698201
Afa34	F: GCAATAGCTCTGCAGGTGTG R: TGCTTGCTGGGATTTGTTCC	60.0 59.8	(AAAC) ₈	7	4	233	209–227	KT698202
Afa35	F: ACCCTGCCAGATCTCTTGTC R: GCCCATTTTCTAAAGAAGATGCC	59.9 59.7	(AGAC) ₁₀ (AGAA) ₇	7	7	166	134–162	KT698203
Afa36	F: TCCTGGCTACTTTATGACACC R: AGACATCTGCAGGGATTGTC	57.9 59.0	(AGAA) ₁₉	3	2	250	266–277	KT698204

Melting temperature (T_m); number of individuals genotyped (n); number of alleles observed (A). All forward primers were 6FAM fluoro-labelled

a powerful biomonitoring tool particularly suitable for rare and elusive species or species sensitive to disturbance.

The Bean Goose (*Anser fabalis*) is a migratory species breeding in the taiga zone or open tundra of northern Eurasia, and Scandinavian birds winter mainly from southern Sweden to the Netherlands (Madsen et al. 1999). The global population appears to be decreasing, and the Bean Goose is currently listed as vulnerable on the Norwegian Red List (Henriksen and Hilmo 2015). As the Bean Goose is very shy and difficult to observe during the breeding season, DNA analyses of non-invasively collected molted feathers may be a valuable complement to traditional field-based approaches for counting and tracking Bean Geese. In this study we applied high-throughput sequencing to identify novel Bean Goose microsatellites. To facilitate genetic monitoring based on non-invasive sampling of molted feathers, we targeted microsatellite loci with short (<250 bp) amplicons, as amplification success

and allele-dropout are affected by amplicon size when genotyping DNA from non-invasively collected samples (Broquet et al. 2007). Furthermore, we developed two multiplex PCR assays for 12 loci and evaluated these assays in a Taiga Bean Goose (*A. fabalis fabalis*) population in central Norway. Finally, these two assays were applied to test for cross-species amplification success in four other *Anser* species.

Samples collected in central and northern Norway from seven Bean Geese were sent as blood stored in lysis buffer and DNA (extracted from feathers) eluted in tris–EDTA buffer to Ecogenics GmbH (Balgach, Switzerland) to produce an enriched microsatellite library and to test for polymorphism. In brief, size-selected fragments from genomic DNA were enriched for simple sequence repeat (SSR) content using magnetic streptavidin beads and biotin-labeled CT and GT repeat oligonucleotides. The SSR-enriched library was analysed on a MiSeq platform

Table 2 Multiplex microsatellite genotyping and sexing of the Bean Goose (*Anser fabalis*)

Locus	MP set	Fluoro-label	Final primer concentration (μM)	Repeat type	<i>n</i>	<i>A</i>	Allele size range (bp)	<i>H_O</i>	<i>H_E</i>	<i>P_{HWE}</i>	<i>F_{NULL}</i>
Afa05	A	6FAM	0.20	Tetra	31	6	176–180	0.74	0.77	0.81	0.01
Afa13	A	NED	0.08	Tetra	31	7	207–231	0.74	0.82	0.08	0.04
Afa15	A	6FAM	0.12	Tetra	31	7	227–251	0.74	0.72	0.52	−0.02
Afa25	A	PET	0.12	Tetra	31	7	220–248	0.84	0.82	0.25	−0.02
Afa30	A	NED	0.08	Tetra	31	10	116–180	0.87	0.84	0.60	−0.03
Afa33	A	6FAM	0.06	Tetra	31	4	91–103	0.58	0.58	0.74	0.00
Afa34	A	VIC	0.04	Tetra	31	5	209–229	0.71	0.73	0.85	0.00
Afa02	B	NED	0.08	Tetra	31	9	190–234	0.77	0.83	0.22	0.03
Afa17	B	6FAM	0.12	Tetra	31	9	217–261	0.84	0.85	0.26	0.00
Afa18	B	PET	0.12	Tetra	31	9	183–227	0.62	0.73	0.04	0.06
Afa19	B	6FAM	0.12	Tetra	31	4	171–183	0.58	0.54	0.71	−0.03
Afa35	B	VIC	0.06	Tetra	31	9	136–192	0.81	0.79	0.53	−0.02
Z-002A	B	HEX	0.08	n/a	14M	1	221	0	0	n/a	
					17F	2	209–221	1	1	n/a	

Number of individuals genotyped (*n*); number of different alleles observed (*A*); observed heterozygosity (*H_O*); expected heterozygosity (*H_E*); probability of deviation from Hardy–Weinberg equilibrium (*P_{HWE}*); estimated frequency of null-alleles according to the Brookfield 1 method implemented in Micro-Checker (van Oosterhout et al. 2004) (*F_{NULL}*); Z-002A (Dawson 2007) were used to determine the sex of the individuals; *M* male, *F* female; *n/a* not applicable

(Illumina, Inc., San Diego, CA, USA) using the Nano 2×250 v2 format. After assembly, 5307 contigs or singlets contained a microsatellite insert with a tetra- or a trinucleotide of at least six repeat units or a dinucleotide of at least ten repeat units. Suitable primer design was possible in 2305 microsatellite candidates, of which 36 were tested for polymorphism. These 36 loci were selected because they yielded amplicons of fewer than 250 base pairs, which facilitates analysis of non-invasively collected samples. Selection was then performed based on genetic variation, single-PCR performance, and peak interpretation, and multiplex PCR was developed through tests of different combinations of markers and adjustments of primer concentrations for the final subsets.

Microsatellite loci were PCR-amplified with fluorescent-labeled forward primers. Single PCRs were performed in a final volume of 10 μL, with a protocol consisting of 95 °C for 15 min, 35 cycles of 95 °C for 30 s, 57 °C for 45 s, 72 °C for 45 s, and a final extension step of 72 °C for 30 min. Multiplexing was performed with a Multiplex PCR Plus Kit (QIAGEN, Hilden, Germany), following the manufacturer's protocol but using an 8.4-μL reaction volume. PCR products (0.8 μL) were mixed with GeneScan 500 LIZ (Applied Biosystems, Foster City, CA, USA) size standard (0.14 μL) and Hi-Di formamide (6.16 μL). Alleles were separated using capillary electrophoresis on an ABI 3130xl Genetic Analyzer and sizes assigned using GeneMapper software (Applied Biosystems).

For genotyping, we used sample types relevant for genetic monitoring and conservation genetics, i.e. blood

from two individuals, plucked feathers from two individuals, and molted feathers from 27 individuals, all collected in the years 1990–2014 in central Norway. Cross-species amplification of the markers was tested by including samples (blood or tissue) from four other *Anser* species. Genomic DNA was extracted from blood using a manual commercial kit (QIAGEN DNeasy Blood & Tissue Kit), following the manufacturer's protocol. DNA from feathers and tissue was extracted using an automated system (Maxwell® 16 Research Instrument; Promega Corp., Madison, WI, USA) and the Maxwell 16 tissue DNA purification kit, following the manufacturer's protocol.

The mean number of alleles, observed and expected heterozygosities, deviation from Hardy–Weinberg equilibrium, and linkage disequilibrium were estimated using Arlequin version 3.5.1.2 (Excoffier and Lischer 2010). A Bonferroni correction for multiple statistical tests (Rice 1989) was applied to linkage disequilibrium *p* values. Null-allele frequencies were estimated with Micro-Checker ver. 2.2.3 (van Oosterhout et al. 2004), and the probability of identity was calculated using GenAlEx 6.501 (Peakall and Smouse 2012).

PCR analysis of three to seven individuals was conducted for the 36 microsatellite loci, and allelic variation was detected in 34 of those (Table 1). Scoring both PCR performance and allele interpretation (results not shown), we selected 12 loci for multiplex PCR development. The two final multiplex PCR assays were typed in 31 presumably unrelated Bean Geese [17 females and 14 males; sex determined with the Z-002A marker (Dawson 2007)]. The

Table 3 Cross-species amplification success of 12 Bean Goose microsatellite loci

Locus	<i>Anser anser</i> ^a (n = 5)	<i>Anser albifrons</i> ^b (n = 2)	<i>Anser brachyrhynchus</i> ^a (n = 5)	<i>Anser erythropus</i> ^b (n = 3)
Afa05	3 (192–200)	3 (188–198)	5 (184–200)	3 (188–200)
Afa13	4 (211–223)	2 (223–227)	5 (211–235)	4 (211–231)
Afa15	3 (219–235)	3 (235–243)	4 (223–239)	2 (243–247)
Afa25	3 (232–248)	4 (232–244)	5 (232–248)	3 (240–252)
Afa30	3 (128–140)	3 (128–140)	3 (128–136)	3 (132–140)
Afa33	1 (99)	2 (91–99)	3 (91–99)	2 (95–99)
Afa34	1 (209)	2 (209–229)	4 (209–237)	3 (209–237)
Afa02	2 (194–202)	3 (198–210)	7 (194–218)	4 (190–218)
Afa17	4 (221–249)	3 (237–245)	3 (241–249)	2 (241–245)
Afa18	4 (199–211)	4 (195–207)	6 (191–227)	3 (191–211)
Afa19	2 (179–183)	2 (175–179)	3 (175–183)	2 (171–179)
Afa35	1 (136)	3 (136–200)	2 (128–136)	2 (132–136)

Number of individuals genotyped (n); shown is the number of alleles, and size range in parentheses

^a Samples obtained from legally hunted birds in Norway

^b Samples provided by the Natural History Museum in Oslo, Norway

12 loci showed a mean of 7.2 alleles per locus (range 4–10; Table 2). Heterozygotes were present in both sexes for all loci, indicating that none are sex-linked. The mean expected heterozygosity was 0.75 (range 0.54–0.85) and mean observed heterozygosity was 0.74 (range 0.58–0.87; Table 1). After correcting for multiple tests, no pairs of loci displayed significant linkage disequilibrium. There was no indication of null-alleles in any locus, but one locus (Afa18) deviated significantly from Hardy–Weinberg equilibrium (Table 2). The combined probability of identity for the 12 loci was 6.5×10^{-13} . Cross-species application of the two multiplex PCR assays revealed that all 12 loci amplified successfully in four other *Anser* species and all loci were polymorphic in the Greater White-fronted Goose (*A. albifrons*), Lesser White-fronted Goose (*A. erythropus*), and Pink-footed Goose (*A. brachyrhynchus*), while three loci were monomorphic in the Greylag Goose (*A. anser*; Table 3). The lower proportion of polymorphic loci in the Greylag Goose is in line with previous studies showing that microsatellite performance decreases with genetic distance (e.g. Primmer et al. 1996), as the Greylag Goose is most distantly related to the Bean Goose among the four *Anser* species tested (Ruokonen et al. 2000). In conclusion, these novel microsatellite loci and their multiplex PCR assays will be useful for family analysis, genetic monitoring, and population genetic analyses of the Bean Goose, as well as for other *Anser* species.

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26977/1-B). We thank Deborah A. Dawson for kindly providing the sex-typing markers. Financial support was received from the County administration in Nord-Trøndelag and the Norwegian Environment Agency. Permissions for Bean Goose catching and tissue sampling were approved by the Norwegian Animal Research Authority (NARA, ID 1775 and ref. 2011/64833) and the Norwegian Environment Agency (ref. 2012/1756).

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